

Fig. 1: Li-ion charge profile

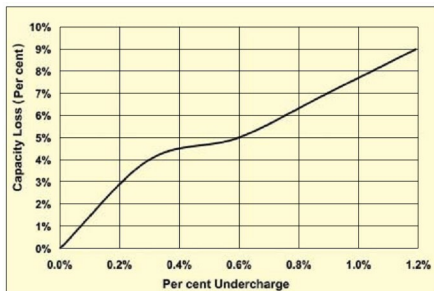


Fig. 2: Capacity loss vs undercharge voltage

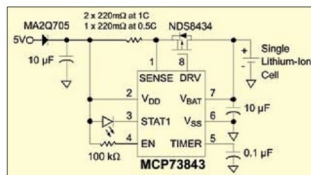


Fig. 3: Typical linear solution

current of 100 mA.

Typically, manufacturers specify the capacity of a battery at a 5-hour rate, $n = 5$. For example, the above-mentioned battery would provide five hours of operating time when discharged at a constant current of 200 mA. In theory, the battery would provide one hour of operating time when discharged at a constant current of 1000 mA. In practice, however, the operating time will be less than one hour due to inefficiencies in the discharge cycle.

So how is energy properly restored to a Li-ion battery? The preferred charge algorithm for Li-ion battery chemistries is a constant-current, constant-voltage algorithm that can be broken up into four stages, namely, trickle charge, constant-current charge, constant-voltage charge and charge termination (refer to Fig. 1):

Stage 1 (trickle charge): Trickle charge is employed to restore charge to deeply depleted cells. When the cell voltage is below approximately 3V, the cell is charged with a constant current of 0.1C maximum.

Stage 2 (constant-current charge): After the cell voltage has risen above the trickle-charge threshold, the charge current is raised to perform constant-current charging. The constant-current charge should be in the 0.2C to 1.0C range. The constant current does not need to be precise and semi-constant current is allowed. Often, in linear chargers, the current is ramped-up as the cell voltage rises in order to minimise heat dissipation in the pass transistor.

Charging at constant current rates above 1C does not reduce the overall charge cycle time and should be avoided. When charging at higher currents, the cell voltage rises more rapidly due to over-voltage in the electrode reactions and the increased voltage across the internal resistance of the cell. The constant-current stage becomes shorter, but the overall charge cycle time is not reduced because the percentage of time in the constant voltage stage increases

the rated capacity in Ah and 'n' is the time in hours at which 'C' is declared.

A battery discharging at a C-rate of '1' will deliver its nominal rated capacity in one hour. For example, if the rated capacity is 1000 mAh, a discharge rate of 1C corresponds to a

in amperes, 'M' is the multiple or fraction of 'C', 'C' is the numerical value of discharge current of 1000 mA. A rate of C/10 corresponds to a discharge